Introduction to Computers

Computer

A computer is a machine that:

- can be programmed to accept data (input)
- process it into useful information (output)
- and store it away (in a secondary storage device) for safekeeping or later reuse.

Components of a Computer

Hardware

People use the power of the computer for some Purpose.

Software

Hardware is the equipment used to perform the necessary computations
and includes the central processing unit (CPU), monitor, keyboard, mouse, printer,
and speakers.

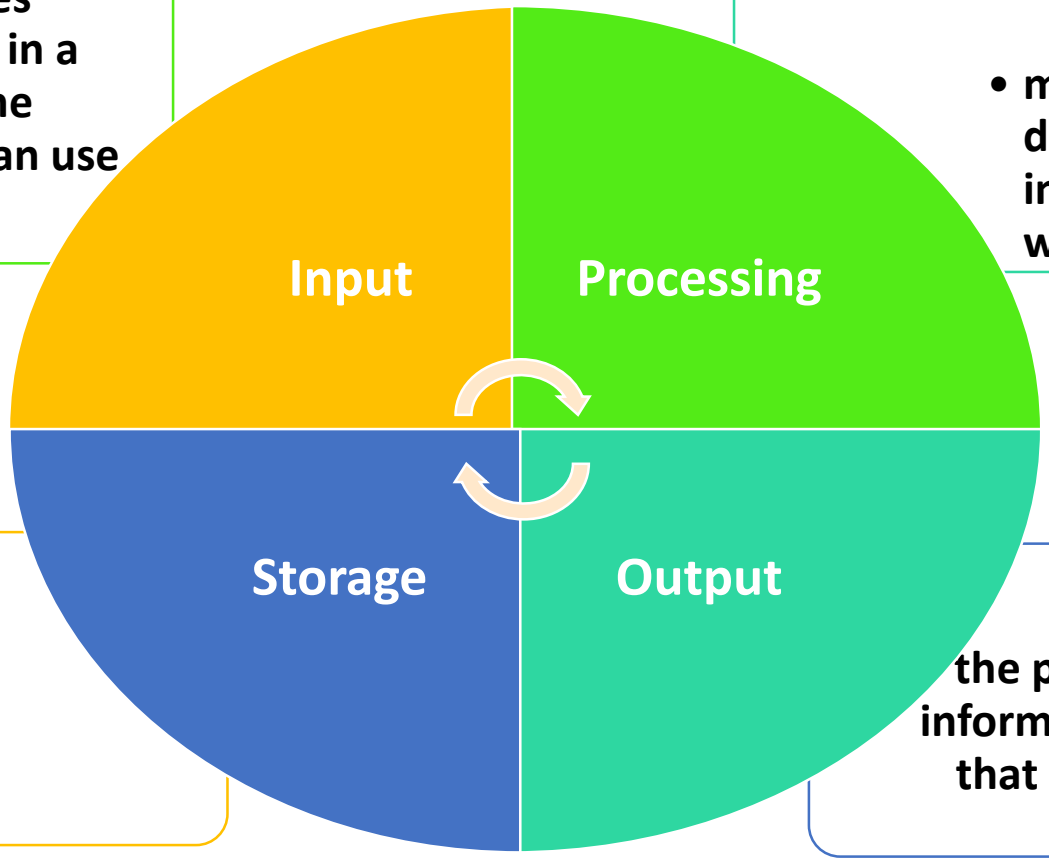
People

Software consists of the **programs** that enable us to solve problems
with a computer by providing it with lists of instructions to perform.

Hardware

- Input devices accept data in a form that the computer can use

- manipulates input data into the information people want.



- store data and programs outside the computer

the processed data-information in a form that people can use

Software:

The planned, step-by-step instructions required to turn data into information.

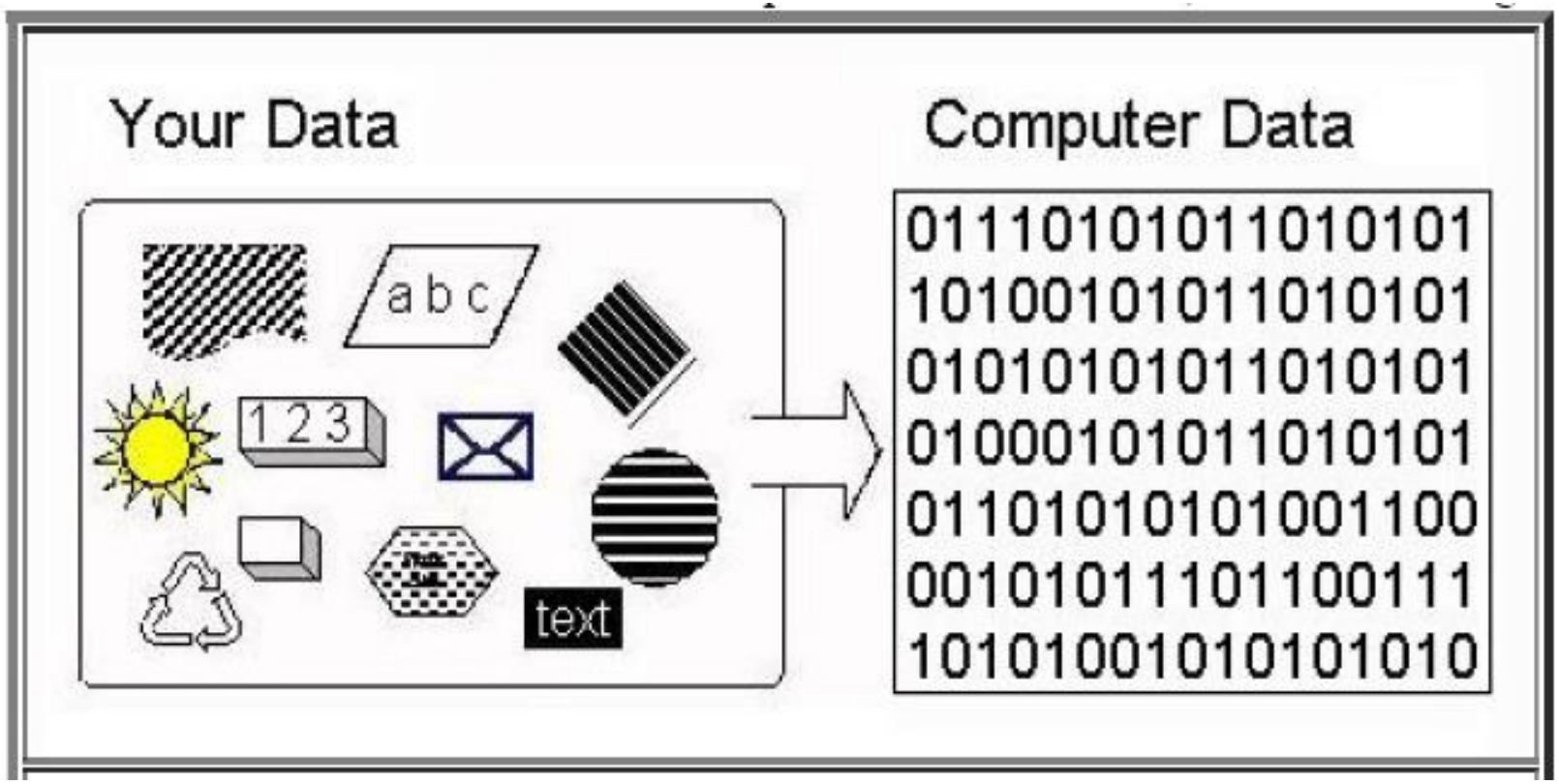
System Software

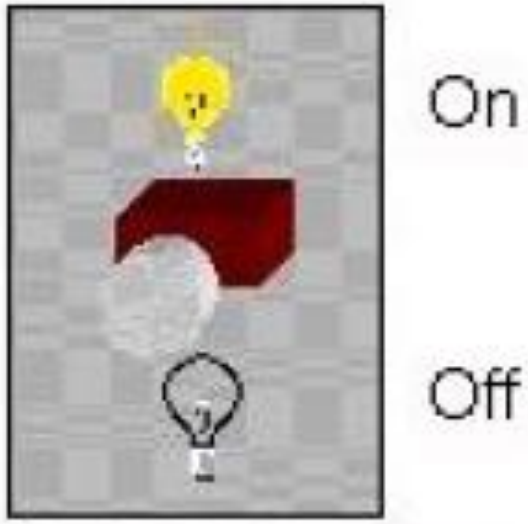
A subset of system software is an operating system, the underlying software found on all computers

Application Software

Software that is applied, can be used to solve a particular problem or to perform a particular task

Representing Data

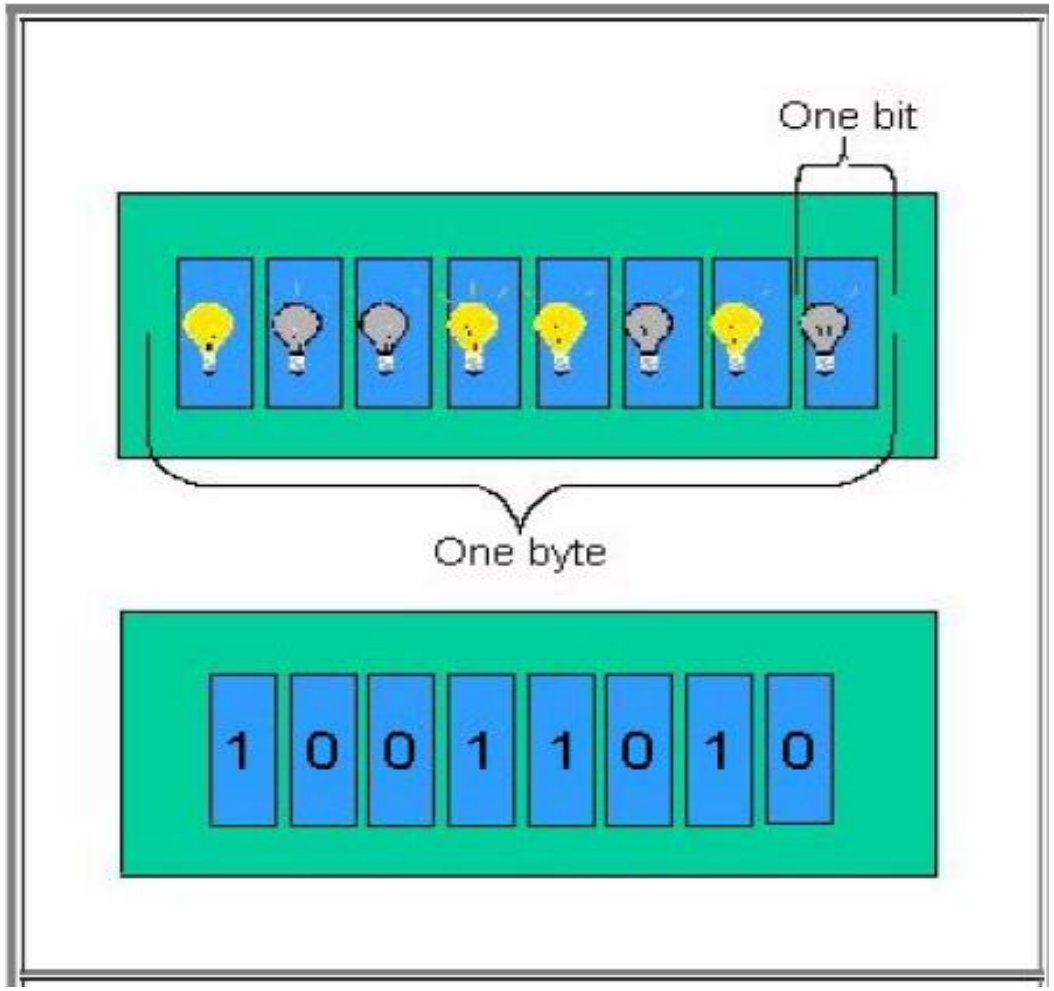




A Switch

Computer manufacturers express the capacity of memory and storage in terms of the number of bytes it can hold

A single byte can represent many different kinds of data. What data it actually represents depends on how the computer uses the byte.



Information in a Byte

- **Characters as a Byte: ASCII**
- **Picture and Graphic Data: grid of pixels (JPEG and PNG)**
- **Sound Data As Bytes: WAV**
- **Program Data as Bytes**

How Does The Computer Know What a Byte Represents?

Evolution of Computers

Progression in hardware representation of a bit of data

Progression of the ease of use of computers

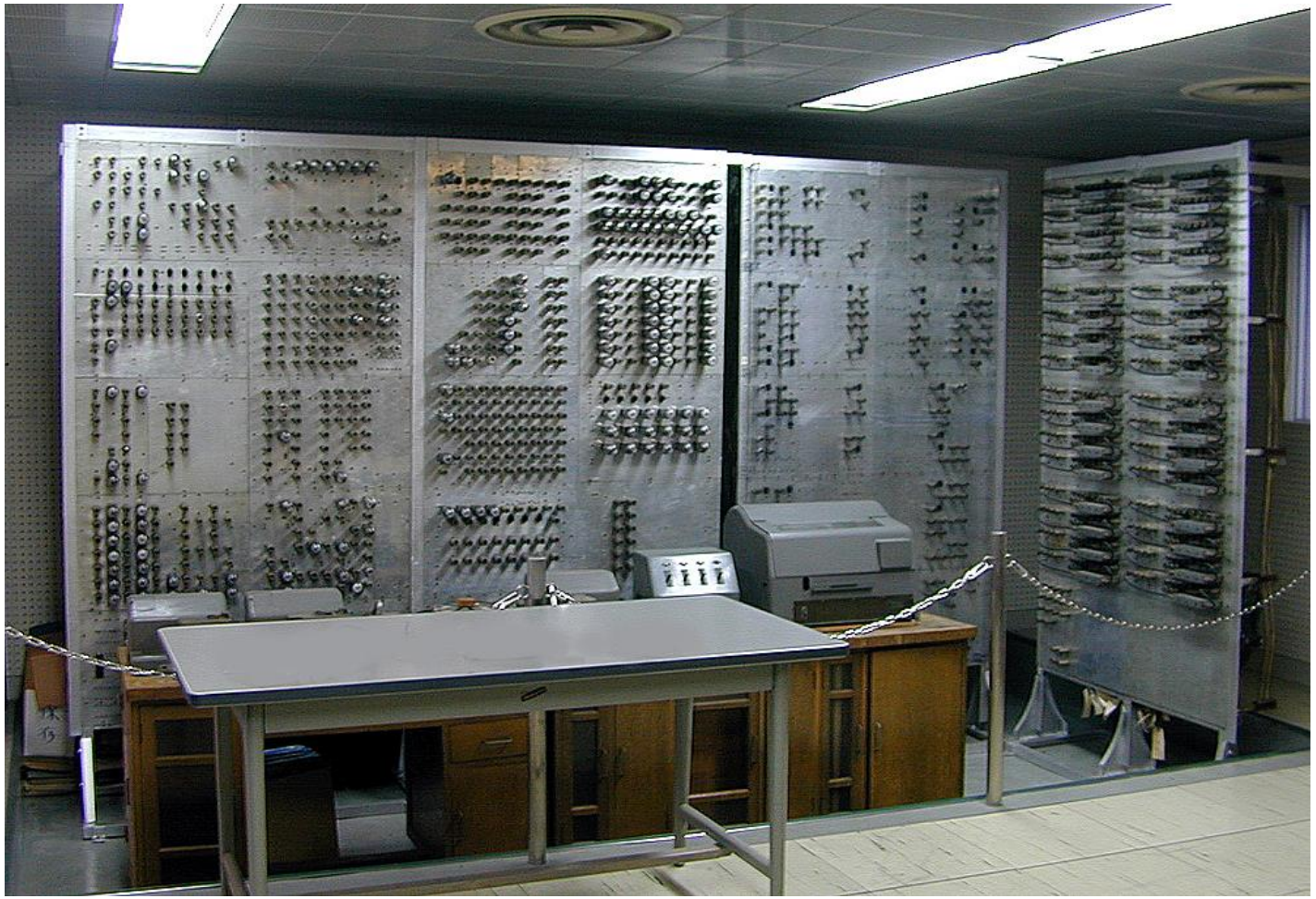
Progression In Hardware Representation

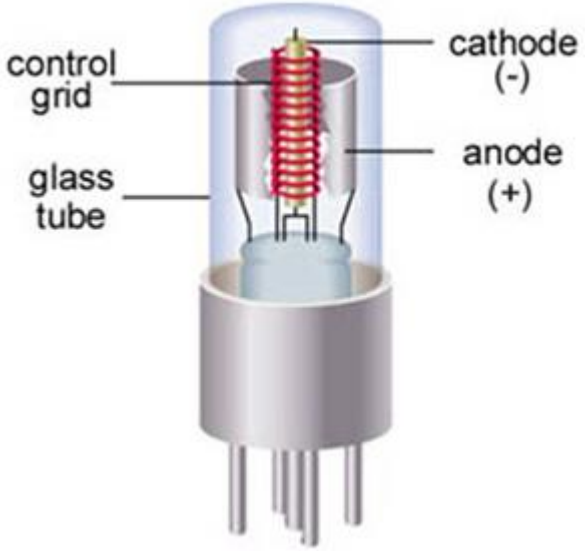
Vacuum Tubes (1950s) - one bit on the size of a thumb; UNIVAC and ENIAC

Transistors (1950s and 1960s) - one bit on the size of a fingernail;

Integrated Circuits (1960s and 70s) - thousands of bits on the size of a hand

Silicon computer chips (1970s and on) - millions of bits on the size of a finger nail.



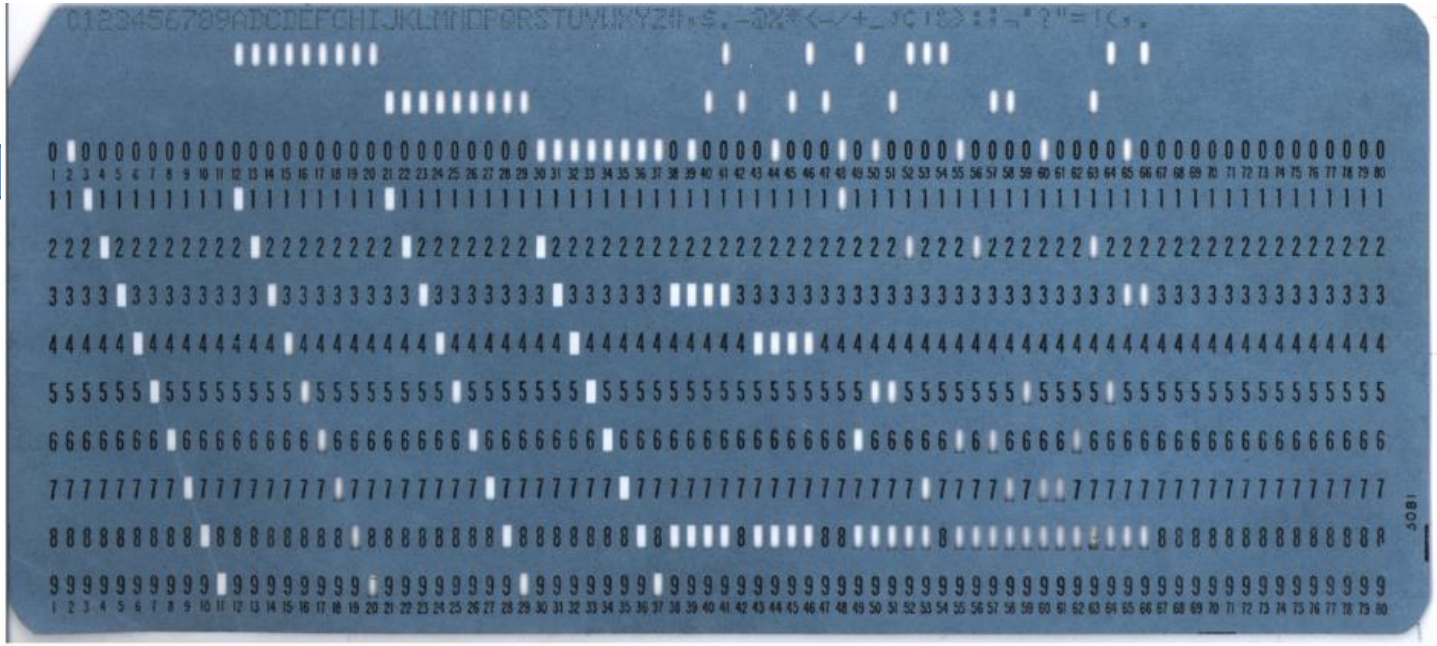

control grid
glass tube
cathode (-)
anode (+)

Progression of the ease of use of computers

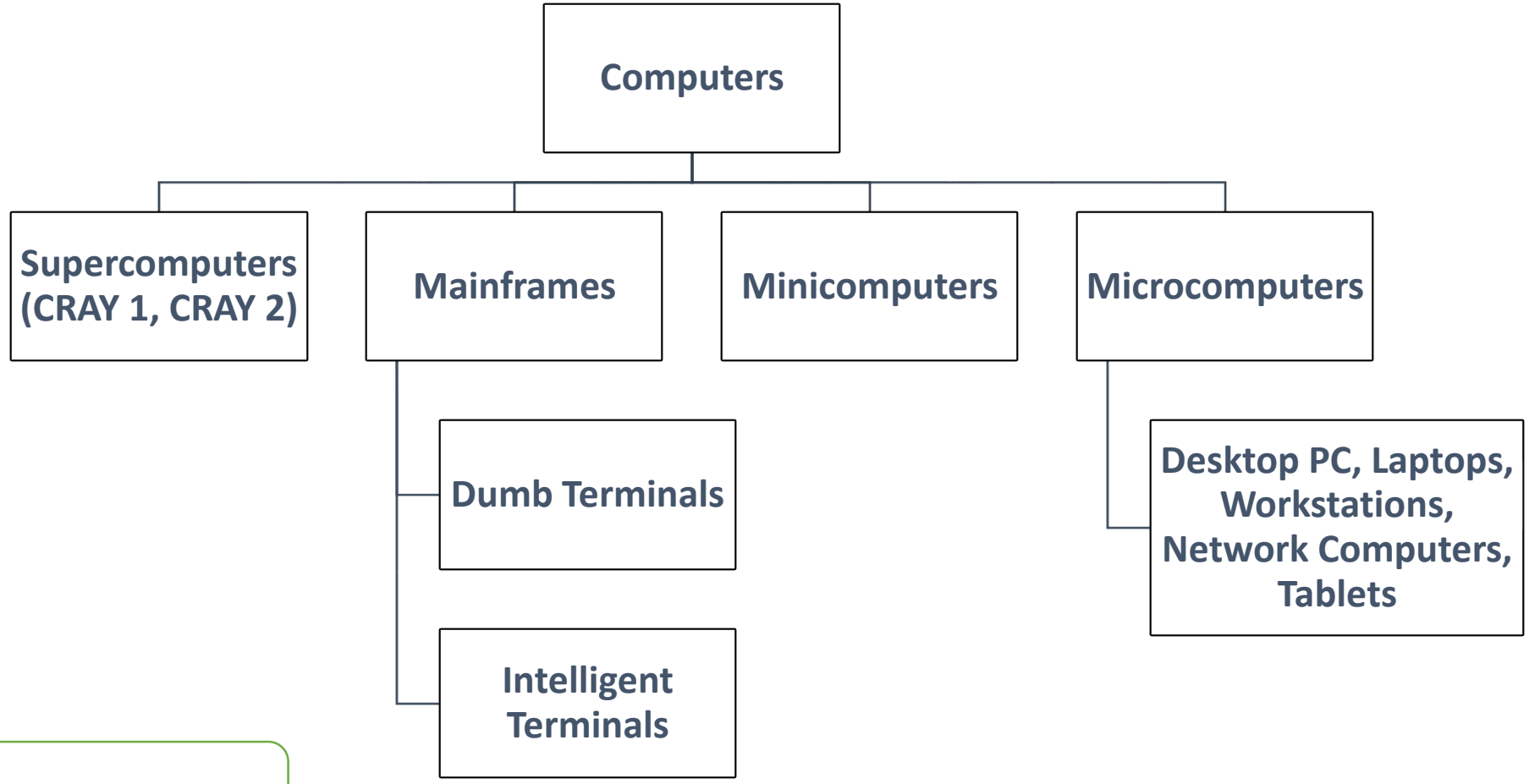
- Almost impossible to use except by very patient geniuses (1950s);
- Programmable by highly trained people only (1960s and 1970s);
- Usable by just about anyone (1980s and on).

no operating systems

Batch Processing System



Classification of Computers



Pratyush

Applications

- Government
- Traffic Control
- Legal System
- Retail Business
- Sports
- Music
- Hospitals etc..